

Lead Fluid Products MODBUS Communication Technology Standards

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Prepared by: Liu Yang Date: 2022-04-09

Reviewed by: Yuan Jichao Date: 2022-04-09

Approved by: Zhang Yanfeng Date: 2022-04-09

Modify Record

No.	Version No.	Date	Prepared	Adjust content
1	The 11th revision of the first edition			
2	The 12th revision of the first edition	2017-07-3	Liu Yang	Add mode description and example
3	The 13th revision of the first edition	2018-04-17	Yuan Jichao	Modify BT-SV2 working mode
4	The 14th revision of the first edition	2020-12-8	Yuan Jichao	Add remarks and modify BT-L and BT-F communication protocols
5	The 15th revision of the first edition	2022-04-09	Shi Tao	Supplementary BT-SV3, BT-F V3
6	The 16th revision of the first edition	2025-09-04	Liang Yilin	Revise the unclear or incorrect descriptions in the peristaltic pump and syringe pump sections. Separate the Syringe Pump V3 and create an independent table for it. Correct the range of certain

				registers in the S series peristaltic pumps.

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1. Introduction

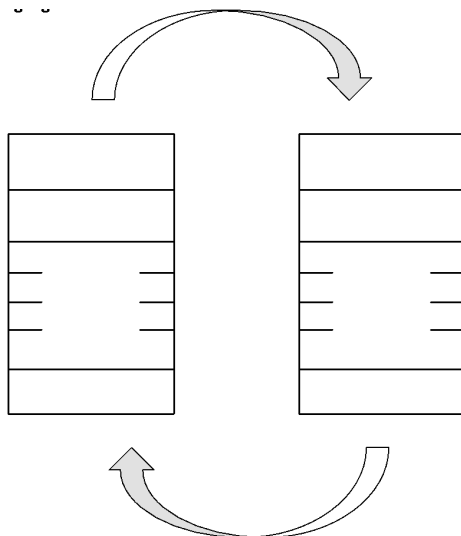
The communication hardware interface of the LEADFLUID series products adopts RS485 and supports the MODBUS protocol. Through this protocol, it is convenient to form centralized monitoring and distributed control with various industrial equipment, and support various industrial controllers (including human-machine interfaces, industrial computers, computers, PLC, etc.). The program adopts modular design which is stable and reliable. This Modbus communication protocol stack includes two layers: the Modbus application layer protocol. Network layer. Our company's products are all transmitted in **RTU** mode.

The currently supported instructions include (Red font instructions are used by the pump)

Function Code	Command Text	Description
0x01	Read coil state	Get the current state of a group of logic coils (ON/OFF)
0x02	Read input state	Get the current state of a group of switch inputs (ON/OFF)
0x03	Read holding register	Get the current binary value in one or more holding registers
0x04	Read input register	Get the current binary value in one or more input registers
0x05	Write single coil	Force the on-off state of a logic coil
0x06	Write a single register	Load the specific binary value into a holding register
0x0F	Write multiple coils	Force the on-off of a series of continuous logic coils
0x10	Write multiple registers	Load specific binary values into a series of continuous holding registers
0x11	Report slave node ID	Enables the host to judge the type of addressing slave and the state of the slave operation indicator
0x17	Read/write multiple registers	Read and write multiple holding registers at the same time

2. Modbus Protocol

2.1. Modbus Protocol Model



2.2. Format of Each Byte

Coding system: 8-bit binary, hexadecimal 0-9, A-F

Data bit: 1 start bit

8-bit data, low bit sent first

Even check 1 bit

Stop bit 1 bit

Error check area: cyclic redundancy check (CRC)

2.3. Information Frame Format

At the beginning of the message, a rest time of at least 3.5 characters is required. According to the baud rate used, it is easy to calculate the rest time (such as T1-T2-T3-T4 in the figure below). The data of the next area is the device address. The characters allowed in each zone are 0-9, A-F in hexadecimal. The equipment on the network continuously monitors the information on the network, including the rest time. When receiving the first address data, each device immediately decodes it to determine whether it is its own address. After the last word symbol is sent, there is also a rest time of 3.5 characters before a new message can be sent. The entire message must be sent continuously. If a rest time greater than 1.5 characters occurs during the transmission of frame information, the receiving device refreshes the incomplete information and assumes the next address data. A new message sent immediately after the same message (if there is no rest time of 3.5 characters) will generate an error. This error is caused by the invalid CRC check code of the merged information.

START	ADDRESS	FUNCTION	DATA	CRC CHECK	END
T1-T2-T3-T4	8 BTI S	8 BTI S	n x 8 BTI S	16 BIT S	T1-T2-T3-T4

RTU information frame

2.3.1. Address Setting

The information address includes 8 bits (RTU), the effective address range of slave device is 0-247, (decimal), and the address range of each slave device is 1-247. The master puts the slave address into the address area of the information frame and addresses the slave. When the slave responds, put its own address into the address area of the response information to let the host recognize the slave address that has responded. The address 0 is the broadcast address, which can be recognized by all slaves. When the Modbus protocol is used for advanced networks, broadcast or its alternative is not allowed. For example, Modbus+uses a token loop to automatically update the shared database.

2.3.2. Function Code Setting

The information frame function code includes 8 characters (RTU). Valid codes range from 1-225 (decimal). Some codes are applicable to all types of controllers, while some codes are only applicable to some types of controllers. There is some code left for future use. When the host sends information to the clause, the function code explains the action to be performed to the slave. For example, read the ON/OFF state of a group of discrete coils or input signals, read the data of a group of registers, read the diagnostic state of the slave, write the coils (or registers), and allow the program in the slave to be cut down, recorded, and confirmed. When the slave responds to the host, the function code can indicate that the slave responds normally or has an error (that

is, abnormal response). When the slave responds normally, the clause simply returns the original function code. In case of abnormal response, the slave returns a code equal to the original code and sets the most significant bit to "1". For example, if the host requests to read a group of holding registers from the host, the function code of sending information is:

0000 0011 (hex 03)

If the slave receives the requested action information correctly, it returns the same code value as the normal response. When an error is found, an abnormal sound message is returned:

1000 0011 (hex 83)

The slave modifies the function code. In addition, a special code is put into the data area of the response information to tell the host the error type and the reason for abnormal response. The application program of the host device is responsible for handling the abnormal response. The typical processing process is that the host sends the test and diagnosis of the information to the slave and notifies the operator.

2.3.3. Content of Data Area

The data area has two hexadecimal data bits and the data range is 00-FF (hexadecimal). According to the mode of network serial transmission, the data area can be composed of a pair of ASCII characters or a RTU character. The information data sent by the master to the slave device includes the request actions specified in the master function code executed by the slave, such as the address of the discrete register, the number of processing objects, and the actual number of data bytes. For example, if the host requests to read a set of registers from the machine (function code 03), the data specifies the starting address of the register and the number of registers. For another example, the host needs to write a set of registers in a slave (the function code is 10H). This data area specifies the starting address to be written to the register area, the number of registers, the number of bytes of data, and the data to be written to the register. If there is no error, the response information from the slave to the host contains the request data. If there is an error, there is an abnormal code in the data, and the host can judge and make the next action. The length of the data area can be "zero" to represent some kind of information, such as the communication event record (function code OBH) of the master request - slave response. At this time, the slave does not need other additional information, and the function code only specifies the action.

2.3.4. Error Verification

The standard Modbus bus has two types of error checking methods. The content of the error checking area is filled according to the error checking method used. The error check code is a 16-bit value and two 8-bit bytes. The error check value is the CRC check result of the message content. CRC check information frame is the last data. The obtained check code is sent to the low byte first, and then to the high byte, so the high byte of CRC code is the last transmitted information.

2.4. Serial Transmission of Information

In the information transmitted on the standard Modbus, each character or byte is transmitted in the order from left to right: least significant bit: (LSB) most significant bit: (MSB)

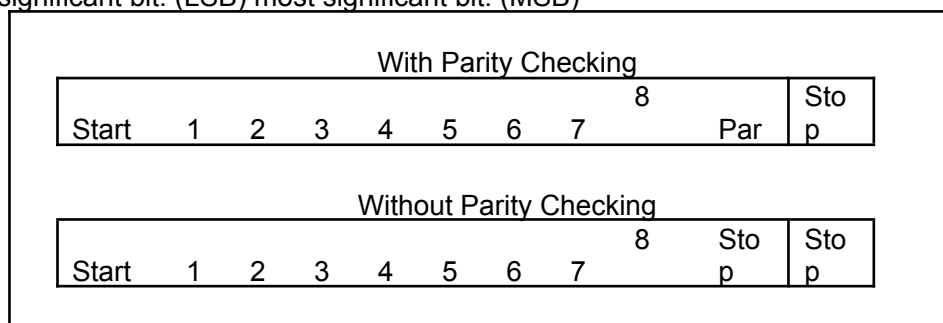


Figure 6 RTU Bit Order

3. Instructions for Using

- 1) During the communication debugging process, first select a single word input register for testing, such as temperature and observe whether the read data matches the display to confirm whether communication is normal. If the data is consistently incorrect, you can try swapping the wiring sequence between A and B.
- 2) The communication state of the LCD display is determined by the transmission and reception of communication data and is not physically connected.
- 3) In this technical standard, if there are no special circumstances, it is recommended to use the default

byte order CDAB (original communication mode: computer mode) for registers with long integers and floating-point numbers. In byte order ABCD (original communication mode: PLC mode), to ensure data correctness, only one instruction is allowed to read and write to a long integer or single precision floating-point variable at a time. In this mode, if two or more single/double word registers are read and written continuously, error data will be generated.

- 4) When communicating with devices, there should be a delay of more than 50ms between each instruction frame to ensure the integrity of communication data. Otherwise, it may cause data errors and communication abnormalities, especially in PLC programming and computer programming. Each instruction frame must be manually added
- 5) When writing registers for flow and liquid volume, it is important to first write the unit before writing the value, otherwise it may cause unit abnormalities and prevent startup
- 6) Modifying the pump number, communication rate, and check bit all require restarting the device to take effect

4. Parameters and Addresses of S Series Peristaltic Pump

Products: BQ50S, BQ80S, BT50S, BT102S, BT100S, BT300S, BT600S, BT101S, BT301S, BT601S, WT300S, WT600S, WT600S-65, WG600S, BT100S-1, BT103S

Mode: RTU

Address: 1-247 settable

Communication rate: 9600

Data bit: 8

Check bit: even check (EVEN)

Stop bit: 1

4.1. Input Register (Not Storable)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	Remarks
1	1000	Reserved	Nothing		Unsigned Short int (2 bytes)	
2	1001	Value of rotating speed timer	The setting that determines current rotating speed	200-65535	Unsigned Short int (2 bytes)	
3	1002	Subdivide number	Step numbers for 1 round	BQ-S:6400 BT-S V2:6400 BT-S:10000	Unsigned Short int (2 bytes)	WT series is not displayed
4	1003	Analog speed control	Speed setting controlled by external analog signal	BQ-S:1-800 BT100S/BT101S/BT100S-1: 1-1500 BT300S/BT301S:1-3500 BT600S/BT601S:1-6000 WT300S:30-350 WT600S/WG600S:30-600	Unsigned Short int (2 bytes)	
5	1018	Manufacturer information	Display manufacturer information	"Lead Fluid"	Unsigned Char (10 bytes)	

6	1023	Product information	Display product information	"BT100S" "BT300S" "BT600S"	Unsigned Char (10 bytes)	
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4.2. Holding Register (Not Storable)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	Remarks
1	3000	Key value	Modify key value	0-8	Unsigned Short int (2 bytes)	
2	3001	Easy dispensing state	Easy dispensing is on or not	Normal: 0 In dispensing: 1	Unsigned Short int (2 bytes)	Suitable for BT-S series
3	3002	Current time dispensing state	Check if the time dispensing running	Normal: 0 Dispensing: 1	Unsigned Short int (2 bytes)	Suitable for BT-S V2 series, BQ-S series, WT-S series, WG-S series. BT-S V2 series requires working mode set timing mode

4.3. Holding Register (Power Down Can Store EEPROM)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	Remarks
1	3100	Rotating Speed	Adjust rotating speed	BQ-S:1-800 BT100S/BT101S:1-1500 BT300S/BT301S:1-3500 BT600S/BT601S:1-6000 WT300S:30-350 WT600S/WG600S:30-600	Unsigned Short int (2 bytes)	Except for the WT series, the value of/10 for other series is the actual speed, corresponding to the following: BQ-S:0.1-80 BT100S/BT101S/WT300S:0.1-150 BT300S/BT301S:0.1-350 BT600S/BT601S:0.1-600
2	3101	Direction state	Set rotation direction	Clockwise: 0 Anticlockwise: 1	Unsigned Short int (2 bytes)	
3	3102	Running state	Start/stop pump	Stop: 0 Start: 1	Unsigned Short int (2 bytes)	
4	3103	Full speed state	Display full speed state	Normal: 0 Full speed: 1	Unsigned Short int (2 bytes)	
5	3104	Control mode	Set control mode	Internal: 0 External: 1 Footswitch: 2 Level: 3	Unsigned Short int (2 bytes)	

				Internal: 0 External: 1 Footswitch: 2 Level: 3 Communicate : 4		Only suitable for BT101S/BT301S /BT601S/WG600S
				Internal: 0 External: 1 Dispense: 2 Level 1: 3 Level 2: 4		Suitable for BT100S V2 /BT300S V2 /BT600S V2 /BT100S-1 V2
6	3105	Easy dispensing volume	Set easy dispensing volume for one micro step	0-4294967295	Unsigned Long int(4bytes)	
7	3107	Slave device address	Set Slave device address	1-99	Unsigned Short int (2 bytes)	
8	3108	MODBUS mode	Switch MODBUS byte order	CDAB(Original computer mode):0 ABCD(Original PLC mode): 1	Unsigned Short int (2 bytes)	
9	3109	Dispensing time	Set dispensing time	1-9999	Unsigned Short int (2 bytes)	Corresponding to 0.1-999. 9
10	3110	Interval time	Interval time between two dispensing processes	1-9999	Unsigned Short int (2 bytes)	Only suitable for BT-S V2, V3 series, corresponding to 0.1-999.9
11	3111	Cycles numbers	Number of cycles	0-999	Unsigned Short int (2 bytes)	Only suitable for BT-S V2, V3 series
12	3112	Reverse angle	Set reverse angle	0-720	Unsigned Short int (2 bytes)	Only suitable for BT-S V2, V3 series
13	3113	Maximum speed	Maximum speed corresponding to analog quantity	1-6000	Unsigned Short int (2 bytes)	Only suitable for BT-S V2, V3 series
14	3114	Language	Set language	0: English 1: Chinese	Unsigned Short int (2 bytes)	Only suitable for BT-S V2, V3 series
15	3115	Beep	Set the Beep to enable or disable	0: Disable 1: Enable	Unsigned Short int (2 bytes)	Only suitable for BT-S V2, V3 series
16	3116	Lock	Set keyboard lock on or off	0: Unlock 1: Lock	Unsigned Short int (2 bytes)	Only suitable for BT-S V2, V3 series
17	3117	Time unit	Unit of operation and interval time	0: Second 1: Minute 2: Hour 3: Day	Unsigned Short int (2 bytes) Running time units do not include	Only suitable for BT-S V2, V3 series Only suitable for BT-S V3 series

					interval time units	
18	3118	Contrast	Set the contrast of LCD	1-20	Unsigned Short int (2 bytes)	Only suitable for BT-S V2, V3 series
19						
20	3124	Acceleration time	Set acceleration time	1-200 corresponds to 0.1S-20.0S	Unsigned Short int (2 bytes)	Only suitable for BT-S V3 series
21	3125	Deceleration time	Set deceleration time	1-200 corresponds to 0.1S-20.0S	Unsigned Short int (2 bytes)	Only suitable for BT-S V3 series
22	3126	Step protection	Disable/enable step protection	0: Disable 1: Enable	Unsigned Short int (2 bytes)	Only suitable for BT-S V3 series
23	3127	Interval time unit	Unit of interval time	0: Second 1: Minute 2: Hour 3: Day	Unsigned Short int (2 bytes)	Only suitable for BT-S V3 series
24	3128	Baud rate	Set baud rate	0: 4800 1: 9600 2: 19200 3: 38400	Unsigned Short int (2 bytes)	Only suitable for BT-S V3 series
25	3129	Baud rate stop bit		1: bit 2: bit	Unsigned Short int (2 bytes)	Only suitable for BT-S V3 series
26	3130	Baud rate verification method		0: No parity 1: Odd check 2: Even check	Unsigned Short int (2 bytes)	Only suitable for BT-S V3 series

5. Parameters and Addresses of L Series Peristaltic Pump

Products: BT100L, BT300L, BT600L, BT101L, BT301L, BT601L

Mode: RTU

Address: 1-247 settable

Communication rate: 4800,9600,19200,38400

Data bit: 8

Check bit: even check (EVEN)

Stop bit: 1

5.1. Input Register (Not Storable)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	Remarks
1	(1000)	Internal temperature	Check the temperature inside the pump	-100-+100 degrees	Signed Short int (2 bytes)	
2	(1001)	Reserved			Unsigned Short int (2 bytes)	
3	(1002)	Rotating speed	Check current rotating speed	BT10XL: 0.1-150 BT30XL: 0.1-350 BT60XL: 0.1-600	Float (4 bytes)	
4	(1004)	Steps has run	Check how	0-4294967295	Unsigned	

		for one cycle dispensing	many steps has run for one cycle dispensing		Long int (4 bytes)	
5	(1006)	Required steps for dispensing and suck-back	Check required steps for dispensing and suck-back	0-4294967295	Unsigned long int (4 bytes)	
6	(1016)	Value of rotating speed timer	The setting that determines current rotating speed	150-65535	Unsigned Short int (2 bytes)	
7	(1018)	Manufacturer information	Display manufacturer information	"Lead Fluid"	Unsigned Char (10 bytes)	
8	(1023)	Product information	Display product information	"BT10XL" "BT30XL" "BT60XL" V3 Series "BT10XLV3" "BT30XLV3" "BT60XLV3"	Unsigned Char (10 bytes)	Not applicable to the V3 series
9	(1028)	Touch panel X-coordinate	Display touch panel X-coordinate		Unsigned Short int (2 bytes)	Not applicable to the V3 series
10	(1029)	Touch panel Y-coordinate	Display touch panel Y-coordinate		Unsigned Short int (2 bytes)	
11	(1030)	Analog speed	Display analog input speed	BT10XL: 1-1500 BT30XL: 1-3500 BT60XL: 1-6000	Unsigned Long int (4 bytes)	Corresponding value BT10XL: 0.1-150 BT30XL: 0.1-350 BT60XL: 0.1-600
12	1032	Accumulated liquid volume Checking	Checking the cumulative liquid volume value		Float (4 bytes)	
13	1034	Accumulated liquid volume unit checking	Checking cumulative liquid volume unit	uL: 1 mL: 2 L: 3	Unsigned Short int (2 bytes)	
14	1036	Maximum speed of pump head	Check the current maximum pump head speed	Unit: rpm	Unsigned Short int (2 bytes)	Suitable for BT-L V1 software version above 1.28 ; BT-L V2 and V3 models
15	1038	Maximum flow rate	Check the current maximum flow rate of the tube	Unit: uL	Float (4 bytes)	Suitable for BT-L V1 software version above 1.28 ; BT-L V2 and V3 models
16	1040	Minimum flow rate	Check the current minimum flow rate of the tube	Unit: uL	Float (4 bytes)	Suitable for BT-L V1 software version above 1.28 ; BT-L V2 and V3 models

5.2. Input Register (Not Storable)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	Remarks
1	(2800)	On time	Check total startup time	0-4294967295 Unit: second	Unsigned Long int (4 bytes)	
2	(2802)	Working time	Check total working time	0-4294967295 Unit: second	Unsigned Long int (4 bytes)	
3	(2804)	Number of powered on times	Check how many times the pump powered on	0-4294967295	Unsigned Long int (4 bytes)	

5.3. Holding Register (Not Storable)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	Remarks
1	3000	Switch checking screen	Switch checking screen	Page 1: 0 Page 2: 1	Unsigned Short int (2 bytes)	

5.4. Holding Register (Power Down Storage FRAM)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	Remarks
1	4000	The leftmost physical value of touch screen	The leftmost physical value of touch screen		Unsigned Long int (4 bytes)	Not applicable to the V3 series
2	4002	The rightmost physical value of touch screen	The rightmost physical value of touch screen		Unsigned Long int (4 bytes)	Not applicable to the V3 series
3	4004	The uppermost physical value of touch screen	The uppermost physical value of touch screen		Unsigned Long int (4 bytes)	Not applicable to the V3 series
4	4006	The down most physical value of touch screen	The down most physical coordinate of touch screen		Unsigned Long int (4 bytes)	Not applicable to the V3 series
5	4008	Group No.	Set Current Group	Time dispensing for 1-5 groups	Unsigned Short int (2 bytes)	
6	4012	Suck-back speed	Set suck-back speed	1-600	Float(4 bytes)	
7	4015	Flow rate in the flow mode	Set the flow rate in the flow mode	0.001-999.9	Float (4 bytes)	
8	4017	Working mode	Set different working modes	Flow: 0 Time: 1	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
9	4018	Keypad tone	Tone on/off	Tone on: 1 Tone off:0	Unsigned Short int (2 bytes)	
10	4019	Parameter locking	Set parameter locking	Unlock: 0 Lock: 1	Unsigned Short int	

					(2 bytes)	
11	4020	Language	Set language	Chinese: 0 English: 1	Unsigned Short int (2 bytes)	
12	4021	Pump head and tube model	Set unified numbering for pump head and tube models	0-102	Unsigned Short int (2 bytes)	
13	4022	Flow rate unit	Set flow rate unit	ul/min 1 ml/min 2 L/min 3	Unsigned Short int (2bytes)	
14	4023	Direction state	Set rotation direction	Clockwise: 0 Anticlockwise: 1	Unsigned Short int (2 bytes)	
15	4024	Full speed state	Indicates whether it is at full speed (cleaning)	Normal: 0 Full speed: 1	Unsigned Short int (2 bytes)	Not applicable to the V3 series
16	4025	Dispensing state	Start command in time dispensing Mode	Stop: 0 Running: 1	Unsigned Short int (2 bytes)	
17	4026	External control mode	Set external control mode	Internal: 0 Foot switch: 1 Current: 2 Voltage: 3	Unsigned Short int (2 bytes)	
18	4027	Reverse angle	Set reverse angle	0-720	Unsigned Short int (2 bytes)	
19	4028	Slave address	Set the slave address for communication	1-247	Unsigned Short int (2 bytes)	
20	4029	Communication baud rate	Set the number of communication baud rate	4800bbs: 0 9600bbs: 1 19200bbs:2 38400bbs: 3	Unsigned Short int (2 bytes)	
21	4030	External control method	Set the method of external control triggering start stop	Pulse: 0 Level: 1	Unsigned Short int (2 bytes)	
22	4031	Infrared enable	Turn infrared function on or off	Disable: 0 Enable: 1	Unsigned Short int (2 bytes)	Not applicable to the V3 series
23	4032	Time dispensing volume	Set the number of microsteps for time dispensing volume	0-4294967295	Unsigned Long int (4 bytes)	Suitable for BT-L V1 series
24	4034	Restore default values	Set default values	Restore: 55 Normal: 170	Unsigned Short int (2 bytes)	
25	4035	Discharge Coefficient		0.1-99999	Float (4 bytes)	
26	4126	Running state	Control start stop in flow mode, check	Stop: 0 Start: 1	Unsigned Short int (2 bytes)	

			whether to run or stop in dispensing mode			
27	4127	MODBUS mode	Switch MODBUS bytes order	CDAB (Original computer mode):0 ABCD (Original PLC mode):1	Unsigned Short int (2 bytes)	
28						
29	4168	Time dispensing data structure	Set time dispensing parameters	(416 8+6 × N) flow 4bytes floating point (4170+6) × N) running time 2 bytes unsigned (4171+6) × N) downtime 2 bytes unsigned (4172+6) × N) 2nd byte unsigned (4173+6) × N) flow unit 2 bytes unsigned (uL/min=1, mL/min=2, L/min=3)	12 * 5 bytes, 30 words, N=0-4, a total of five sets of data	Suitable for BT-L V2,V3 series
30						
31	4256	Communication check bit	Set communication verification method	No parity: 0 Odd check: 1 Even parity check: 2	Unsigned Short int (2 bytes)	
32	4257	LCD backlight	Set the brightness of the LCD backlight	0-100	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
33	4258	Hours of timed start	Set the hourly value for timed start in flow mode	0-999 hour	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
34	4259	Minutes of timed start	Set the minute value for timed start in flow mode	0-59 minute	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
35	4260	Seconds of timed start	Set the second value for timed start in flow mode	0-59 second	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
36	4261	Hours of timed stop	Set the hourly value for timed start in flow mode	0-999 hour	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
37	4262	Minutes of timed stop	Set the minute value for timed start in flow mode	0-59 minute	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series

38	4263	Seconds of timed stop	Set the second value for timed start in flow mode	0-59 second	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
39	4265	Interval time unit	Set interval time units for time dispensing and liquid volume dispensing	Second: 0 Minute: 1 Hour: 2	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
40	4267	Analog speed of 0V	Speed at 0V analog	Value /10 Unit: rpm	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
41	4268	Analog speed of 5V	Speed at 5V analog	Value /10 Unit: rpm	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
42	4269	Analog speed of 4mA	Speed at 4mA analog	Value /10 Unit: rpm	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
43	4270	Analog speed of 20mA	Speed at 20mA analog	Value /10 Unit: rpm	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
44	4271	External control direction control signal	External control signal triggering direction control method	Pulse: 0 Level: 1	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
45	4272	Pulse control signal	Pulse control method	Descending edge: 0 Rising edge: 1	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series
46	4273	Level control signal	Level control method	Low level: 0 High level: 1	Unsigned Short int (2 bytes)	Suitable for BT-L V2, V3 series

5.5. Holding Register (Power Down Storage RAM)

No.	Address	Name	Function	Data Range	Data Type	Remarks
1	4800	Total number of running times	Record cumulative times	0- 4294967295	UnSigned Long int (4 bytes)	
2	4802	Total number of running steps	Record the cumulative steps (equivalent to the cumulative liquid volume)	0- 4294967295	Unsigned Long int (4 bytes)	Clearing it to zero can achieve the resetting of accumulated liquid volume

6. Parameters and Addresses of F Series Peristaltic Pump

Products: BT100F, BT300F, BT600F, BT101F, BT301F, BT601F, BT100F-1, WT300F, WT600F, WG600F

Mode: RTU

Address: 1-247 settable

Communication rate: 4800,9600,19200,38400

Data bit: 8

Check bit: even check (EVEN)

Stop bit: 1

6.1. Input Register (Not Storable)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	Remarks
1	(1000)	Internal temperature	Check the temperature inside the pump	-100-+100 degrees	Signed Short int (2 bytes)	
2	(1001)	Reserved			Unsigned Short int (2 bytes)	
3	(1002)	Rotating speed	Check current rotating speed	BT10XF 0.1-150 BT30XF 0.1-350 BT60XF 0.1-600	Float (4 bytes)	
4	(1004)	Steps has run for one cycle dispensing	Check how many steps has run for one cycle dispensing	0-4294967295	Unsigned Long int (4 bytes)	
5	(1006)	Required steps for dispensing and suck-back	Check required steps for dispensing and suck-back	0-4294967295	Unsigned long int (4 bytes)	
6	(1008)	Current dispensing time	Check current dispensing time	0-4294967295	Unsigned long int (4 bytes)	
7	(1010)	Current dispensing volume	Check current dispensing volume	0-4294967295	Unsigned Long int (4 bytes)	
8	(1012)	Current dispensing times	Check current dispensing times	0-4294967295	Unsigned Long int (4 bytes)	
9						
10						
11	(1016)	Value of rotating speed timer	The setting that determines current rotating speed	150-65535	Unsigned Short int (2 bytes)	
12	(1018)	Manufacturer information	Display manufacturer information	"LeadFluid"	Unsigned Char (10 bytes)	
13	(1023)	Product information	Display product information	"BT100F" "BT300F" "BT600F" V3 Series: "BT10XFV3" "BT30XFV3" "BT60XFV3"	Unsigned Char (10 bytes)	
14	(1028)	Touch panel X-coordinate	Display touch panel X-coordinate		Unsigned Short int (2 bytes)	Not applicable to the V3 series
15	(1029)	Touch panel Y-coordinate	Display touch panel Y-coordinate		Unsigned Short int (2 bytes)	Not applicable to the V3 series
16	(1030)	Analog speed	Display analog	BT10XF: 1-1500	Unsigned	Corresponding value

		control	input speed	BT30XF: 1-3500 BT60XF: 1-6000	Long int (4 bytes)	BT10XF: 0.1-150 BT30XF: 0.1-350 BT60XF: 0.1-600
17	1032	Accumulated liquid volume checking	Checking the cumulative liquid volume value		Float (4 bytes)	
18	1034	Accumulated liquid volume unit checking	Checking cumulative liquid volume unit	uL: 1 mL: 2 L: 3	Unsigned Short int (2 bytes)	
19	1036	Maximum speed of pump head	Check the current maximum pump head speed	Unit: rpm	Unsigned Short int (2 bytes)	Suitable for BT-F V1 software version 1.28 or above BT-L V2, V3
20	1038	maximum flow rate	Check the current maximum flow rate of the tube	Unit: uL	Float (4 bytes)	Suitable for BT-F V1 software version 1.28 or above BT-L V2, V3
21	1040	Minimum flow rate	Check the current minimum flow rate of the tube	Unit: uL	Float (4 bytes)	Suitable for BT-F V1 software version 1.28 or above BT-L V2, V3

6.2. Input Register (Not Storable)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	Remarks
1	(2800)	On time	Check total startup time	0-4294967295 Unit: second	Unsigned Long int (4 bytes)	
2	(2802)	Working time	Check total working time	0-4294967295 Unit: second	Unsigned Long int (4 bytes)	
3	(2804)	Number of powered on times	Check how many times the pump powered on	0-4294967295	Unsigned Long int (4 bytes)	

6.3. Holding Register (Not Storable)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	Remarks
1	3000	Switch checking screen	Switch checking screen	Page 1: 0 Page 2: 1	Unsigned Short int (2 bytes)	

6.4. Holding Register (Power Down Storage FRAM)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	Remarks
1	4000	The leftmost physical value of touch screen	The leftmost physical value of touch screen		Unsigned Long int (4 bytes)	
2	4002	The rightmost physical value of touch screen	The rightmost physical value of touch screen		Unsigned Long int (4 bytes)	
3	4004	The uppermost physical value	The uppermost physical value		Unsigned Long int	

		of touch screen	of touch screen		(4 bytes)	
4	4006	The down most physical value of touch screen	The down most physical value of touch screen		Unsigned Long int (4 bytes)	
5	4008	Group No.	Set Current Group	Time dispensing for 1-5 groups	Unsigned Short int (2 bytes)	
6	4012	Suck-back speed	Set suck-back speed	1-600	Float(4 bytes)	
7	4015	Flow rate in the flow mode	Set the flow rate in the flow mode	0.001-999.9	Float (4 bytes)	
8	4017	Working mode	Set different working modes	Flow: 0 Liquid volume dispensing: 1 Time dispensing: 2 Copy: 3	Unsigned Short int (2 bytes)	In V2,copy dispensing is a loop dispensing: 3 In V3, copy dispensing is programming mode: 3
9	4018	Keypad tone	Tone on/off	Tone on: 1 Tone off: 0	Unsigned Short int (2 bytes)	
10	4019	Parameter locking	Set parameter locking	Unlock: 0 Lock: 1	Unsigned Short int (2 bytes)	
11	4020	Language	Set language	Chinese: 0 English: 1	Unsigned Short int (2 bytes)	
12	4021	Pump head and tube model	Set unified numbering for pump head and tube models	0-102	Unsigned Short int (2 bytes)	
13	4022	Flow unit	Set flow unit	ul/min: 1 ml/min: 2 L/min: 3	Unsigned Short int (2 bytes)	
14	4023	Direction state	Set rotation direction	Clockwise: 0 Anticlockwise: 1	Unsigned Short int (2 bytes)	
15	4024	Full speed state	Indicates whether it is at full speed (cleaning)	Normal: 0 Full speed: 1	Unsigned Short int (2 bytes)	Not applicable to the V3 series
16	4025	Dispensing state	Start to run in dispensing mode	Stop: 0 Running: 1	Unsigned Short int (2 bytes)	Time dispensing, volume dispensing(cycle dispensing/programming mode) belongs to dispensing mode
17	4026	External control mode	Set external control mode	Internal control: 0 Foot switch: 1 Current: 2 Voltage: 3	Unsigned Short int (2 bytes)	
18	4027	Reverse angle	Set reverse angle	0-720	Unsigned Short int (2 bytes)	

19	4028	Slave address	Set the slave address for communication	1-247	Unsigned Short int (2 bytes)	
20	4029	Communication baud rate	Set the number of communication baud rate	4800bbs 0 9600bbs 1 19200bbs 2 38400bbs 3	Unsigned Short int (2 bytes)	
21	4030	External control method	Set the method of external control triggering start stop	Pulse: 0 Level: 1	Unsigned Short int (2 bytes)	
22	4031	Infrared enable	Turn infrared function on or off	Disable: 0 Enable: 1	Unsigned Short int (2 bytes)	Not applicable to the V3 series
23	4034	Restore default values	Set default values	Restore: 55 Normal: 170	Unsigned Short int (2 bytes)	
24	4035	Discharge Coefficient		0.1-99999	Float (4 bytes)	
25	4126	Running state	Control start stop in flow mode, check whether to run or stop in dispensing mode	Stop: 0 Start: 1	Unsigned Short int (2 bytes)	
26	4127	MODBUS mode	Switch MODBUS bytes order	CDAB (Original computer mode) : 0 ABCD (Original PLC mode) : 1	Unsigned Short int (2 bytes)	
27	4128	Liquid volume dispensing data structure	Set liquid volume dispensing parameters	(4128+8 × N) Flow 4 bytes floating point (4130+8) × N Liquid volume 4 bytes floating point (4132+8) × N Downtime 2 bytes unsigned (4133+8) × N 2nd byte unsigned (4134+8) × N The unit of liquid volume is 2 bytes, unsigned (1 in microliters, 2 in milliliters, and 3 in liters) (4135+8) × N Flow unit 2 bytes unsigned (uL/min=1, mL/min=2, L/min=3)	16*5 bytes 40 words N=0-4, a total of five sets of data	
28	4168	Time dispensing	Set time dispensing	(4168+6 × N) Flow 4 bytes	12 * 5 bytes, 30	

		data structure	parameters	floating point (4170+6) × N) Running time 2 bytes unsigned (4171+6) × N) Downtime 2 bytes unsigned (4172+6) × N) 2nd byte unsigned (4173+6) × N) Flow unit 2 bytes unsigned (uL/min=1, mL/min=2, L/min=3)	words, N=0-4, a total of five sets of data	
29	4198	Copy dispensing data structure	Set replication dispensing parameters	(4198+8) × N) Flow 4 bytes floating point (4200+8 × N) Total liquid volume 4 bytes floating point (4202+8 × N) Downtime 2 bytes unsigned (4203+8 × N) 2nd byte unsigned (4204+8 × N) The unit of total liquid volume is 2 bytes, unsigned (1 in microliters, 2 in milliliters, and 3 in liters) (4205+8 × N) Flow unit 2 bytes unsigned (uL/min=1, mL/min=2, L/min=3)	16*5 bytes 40 words N=0-4, a total of five sets of data	Suitable for BT-F V1 series
30	4256	Communicatio n check bit	Set communication verification method	No parity: 0 Odd check: 1 Even parity check: 2	Unsigned Short int (2 bytes)	
31	4257	LCD backlight	Set the brightness of the LCD backlight	0-100	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
32	4258	Hours of timed start	Set the hourly value for timed start in flow mode	0-999 hour	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
33	4259	Minutes of timed start	Set the minute value for timed start in flow mode	0-59 minute	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
34	4260	Seconds of timed start	Set the second value for timed	0-59 second	Unsigned Short int	Suitable for BT-F V2, V3 series

			start in flow mode		(2 bytes)	
35	4261	Hours of timed stop	Set the hourly value for timed start in flow mode	0-999 hour	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
36	4262	Minutes of timed stop	Set the minute value for timed start in flow mode	0-59 minute	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
37	4263	Seconds of timed stop	Set the second value for timed start in flow mode	0-59 second	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
38	4265	Interval time unit	Set interval time units for time dispensing and liquid volume dispensing	Second: 0 Minute: 1 Hour: 2	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
39	4267	Analog speed of 0V	Speed at 0V analog	Value /10 Unit: rpm	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
40	4268	Analog speed of 5V	Speed at 5V analog	Value /10 Unit: rpm	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
41	4269	Analog speed of 4mA	Speed at 4mA analog	Value /10 Unit: rpm	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
42	4270	Analog speed of 20mA	Speed at 20mA analog	Value /10 Unit: rpm	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
43	4271	External control direction control signal	External control signal triggering direction control method	Pulse: 0 Level: 1	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
44	4272	Pulse control signal	Pulse control method	Descending edge: 0 Rising edge: 1	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
45	4273	Level control signal	Level control method	Low level: 0 High level: 1	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
46	4275	Transfer mode	Flow transfer mode	Flow mode: 0 Time mode: 1	Unsigned Short int (2 bytes)	Suitable for BT-F V3 series
47	4276	Flow unit	The unit of transmitted liquid volume	Volume unit: 0 Weight unit: 1	Unsigned Short int (2 bytes)	Suitable for BT-F V3 series
48	4277	Locked rotor alarm	Locked rotor alarm development	Close: 0 Open: 1	Unsigned Short int (2 bytes)	Suitable for BT-F V3 series
49	4278	Dispensing completion alarm	Set dispensing alarm values	50-100 (APP pushes an alarm when the dispensing progress reaches the set value)	Unsigned Short int (2 bytes)	Suitable for BT-F V3 series

50						
51	4297	Overall steps of circular dispensing	How many steps are there under cyclic dispensing	1-30	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
52	4298	Total number of cycles	The total number of cycles under cyclic dispensing	0-999	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
53	4299	Display the current parameters	Which step's parameters are displayed on the LCD	1-30	Unsigned Short int (2 bytes)	Suitable for BT-F V2, V3 series
54	4300	Programming mode data structure	Set programming mode data parameters	(4300+10 × N) flow 4 bytes floating point (4302+10 × N) liquid volume 4 bytes floating point (4304+10 × N) Unsigned interval of 2 bytes (4305+10 × N) number of cycles 2 bytes unsigned (4306+10 × N) liquid volume unit 2 bytes, unsigned (1 in microliters, 2 in milliliters, and 3 in liters) (4307+10 × N) flow unit 2 bytes unsigned (uL/min=1, mL/min=2, L/min=3) (4308+10 × N) interval unit 2 bytes unsigned (seconds are 0, minutes are 1, and hours are 2) (4309+10 × N) rotation direction 2 bytes unsigned (0 clockwise, 1 counterclockwise)	20 * 30 bytes, 300 words, N=0-29, a total of 30 sets of data	Suitable for BT-F V2, V3 series
55	4600	Time dispensing data structure (Supplementary 4168)	Set time dispensing parameters	(4600+2 × N) time running time unit 2 bytes unsigned (4601+2 × N) downtime unit 2 bytes unsigned (seconds are 0, minutes are 1, and hours are 2)	4 * 5 bytes, 10 words, N=0-4, a total of five sets of data	Suitable for BT-F V2, V3 series

56	4610	Liquid volume dispensing data structure (Supplementary 4128)	Set liquid volume dispensing parameters	(4610+4 × N) running time 4 bytes floating point (4130+4 × N) running time unit 2 bytes unsigned (4132+3) × N) downtime unit 2 bytes unsigned (seconds are 0, minutes are 1, and hours are 2)	8*5 bytes words, N=0-4, a total of five sets of data	Suitable for BT-F V2, V3 series
57	4630	Programming mode data structure (Supplementary 4300)	Set programming mode and parameters	(4630+3) × N) running time 4 bytes floating point (4630+3) × N) running time unit 2 bytes unsigned (seconds are 0, minutes are 1, and hours are 2)	6 * 30 bytes, 90 words, N=0-29, a total of 30 sets of data	Suitable for BT-F V2, V3 series
58						
59						
60						
61						
62						

6.5. Holding Register (Power Down Storage RAM)

No.	Address	Name	Function	Data Range	Data Type	Remarks
1	4800	Total number of running times	Record cumulative times	0- 4294967295	UnSigned Long int (4 bytes)	
2	4802	Total number of running steps	Record the cumulative steps (equivalent to the cumulative liquid volume)	0- 4294967295	Unsigned Long int (4 bytes)	

7. Parameters and Addresses of S Series Gear Pump

Products: CT3000S, CT3001

Mode: RTU

Addresses: 1-247 (User Defined)

Baud Rate: 9600

Data Bits: 8

CRC: Even Parity

Stop Bit: 1

7.1. Input Register (Not Storable)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	
1	1000	Reserved			Unsigned Short int (2 bytes)	
2	1001	Value of rotating speed timer	The setting that determines current rotating speed	99-998	Unsigned Short int (2 bytes)	
3	1003	Analog speed control	Speed setting controlled by external analog signal	300-3000rpm	Unsigned Short int (2 bytes)	
4	1018	Manufacturer information	Display manufacturer information	"LeadFluid"	Unsigned Char (10 bytes)	
5	1023	Product information	Display product information	"CT3000S" "CT3001S"	Unsigned Char (10 bytes)	

7.2. Table 4.2 Holding Register (Not Stored)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	
1	3000	Key value	Modify key value		Unsigned Short int (2 bytes)	
2	3001					
3	3002	Current time dispensing state	Check if the time dispensing running	Stopped: 0 Dispensing: 1	Unsigned Short int (2 bytes)	

7.3. Table 4.3 Holding Register (Power-off Memory EEPROM)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	
1	3100	Rotating Speed	Adjust rotating speed	300-3000rpm	Unsigned Short int (2 bytes)	
2						
3	3102	Running state	Start/stop pump	Stop: 0 Start: 1	Unsigned Short int (2 bytes)	
4	3103	Full speed state	Display full speed state	Normal: 0 Full speed: 1	Unsigned Short int (2 bytes)	
5	3104	Control mode	External, foot switch, or internal control modes	Internal: 0 External: 1 Footswitch: 2 Level: 3	Unsigned Short int (2 bytes)	
6						
7	3107	Slave	Set Slave device	1-247	Unsigned	

		device address	address		Short int (2 bytes)	
8	3108	MODBUS mode	Switch MODBUS mode	Computer: 0 PLC: 1	Unsigned Short int (2 bytes)	
9	3109	Dispensing time	Set dispensing time	1-9999 (0.1-999.9 s)	Unsigned Short int (2 bytes)	

8. Parameters and Addresses of F Series Gear Pump

Products: CT3000F, CT3001F

Mode: RTU

Address: 1-247 settable

Communication rate: 4800, 9600, 19200, 38400

Data bit: 8

CRC: Even Parity

Stop bit: 1

8.1. Table 4.2.1 Input Register (Not Stored)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	
1	(1000)	Internal temperature	Check the temperature inside the pump	-100-+100 degrees	Signed Short int (2 bytes)	
2	(1001)	Reserved			Unsigned Short int (2 bytes)	
3	(1002)	Rotating speed	Check current rotating speed	50-3000rpm	Unsigned Short int (2 bytes)	
4	(1004)	Current running step	Check current running step	0-4294967295	Unsigned Long int (4 bytes)	
5	(1006)	Requisite running steps	Check requisite running steps	0-4294967295	Unsigned long int (4 bytes)	
6	(1008)	Current dispensing time	Check the current dispensing time (seconds)	0-4294967295	Unsigned long int (4 bytes)	
7	(1010)	Current dispensing volume	Check current dispensing volume (μL)	0-4294967295	Unsigned Long int (4 bytes)	
8	(1012)	Current dispensing times	Check current dispensing times	0-4294967295	Unsigned Long int (4 bytes)	
9						
10	(1016)	Value of rotating speed timer	The setting that determines current rotating speed	720-43200	Unsigned Short int (2 bytes)	

11	(1018)	Manufacturer information	Display manufacturer information	"LeadFluid"	Unsigned Char (10 bytes)	
12	(1023)	Product information	Display product information	"CT3000F" "CT3001F"	Unsigned Char (10 bytes)	
13	(1028)	Touch panel X coordinate	Display touch panel the X coordinate		Unsigned Short int (2 bytes)	
14	(1029)	Touch panel Y coordinate	Display touch panel the Y coordinate		Unsigned Short int (2 bytes)	
15	(1030)	Analog speed control	Display	50-3000rpm	Unsigned Long int (4 bytes)	
16	1032	Cumulative volume	Check cumulative volume		Float (4 bytes)	
17	1034	Unit of cumulative volume	Check unit of cumulative volume	μL: 1 mL: 2 L: 3	Unsigned Short int (2 bytes)	
18	(1980)	Error log (20 groups)	Check error log		Unsigned Short int (2 bytes)	

8.2. Input register (not stored)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	
1	(2800)	Total On time	Check cumulative time that the pump powered on (seconds)	0-4294967295	Unsigned Long int (4 bytes)	
2	(2802)	Total running time	Check cumulative running time (seconds)	0-4294967295	Unsigned Long int (4 bytes)	
3	(2804)	Total powered on times	Check how many times the pump powered on	0-4294967295	Unsigned Long int (4 bytes)	

8.3. Holding Register (Not Stored)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	
1	3000	Switch monitoring screen	Switch monitoring screen	Page 1: 0 Page 2: 1	Unsigned Short int (2 bytes)	

8.4. Holding Register (Power-off Memory FRAM)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	
1	4000	The leftmost	The leftmost physical value of		Unsigned Long int	

		physical value of touch screen	touch screen		(4 bytes)	Not applicable to the V3 series
2	4002	The rightmost physical value of touch screen	The rightmost physical value of touch screen		Unsigned Long int (4 bytes)	Not applicable to the V3 series
3	4004	The uppermost physical value of touch screen	The uppermost physical value of touch screen		Unsigned Long int (4 bytes)	Not applicable to the V3 series
4	4006	The down most physical value of touch screen	The down most physical value of touch screen		Unsigned Long int (4 bytes)	Not applicable to the V3 series
5	4008	Dispensing parameter group setting	Set dispensing parameter group	Group 1: 0 Group 2: 1 Group 3: 2 Group 4: 3	Unsigned Short int (2 bytes)	
6						
7	4015	Flow rate	Flow rate setting for flow mode	0.001-999.9	Float (4 bytes)	
8	4017	Working mode	Working mode setting	Flow: 0 Volume: 1 Time: 2 Copy: 3	Unsigned Short int (2 bytes)	
9	4018	Key tone	Tone on/off	Tone on: 1 Tone off: 0	Unsigned Short int (2 bytes)	
10	4019	Parameter lock	Set parameter lock	Unlock: 0 Lock: 1	Unsigned Short int (2 bytes)	
11	4020	Language	Language setting	Chinese: 0 English: 1	Unsigned Short int (2 bytes)	
12	4021	Model of pump head	Pump head setting	MS204: 0 MS209: 7 MS213: 9	Unsigned Short int (2 bytes)	
13	4022	Flow rate unit	Flow rate unit for flow mode	uL/min: 1 mL/min: 2 L/min: 3	Unsigned Short int (2 bytes)	
14						
15	4024	Full speed	Full speeding setting (cleaning)	Normal: 0 Full speed: 1	Unsigned Short int (2 bytes)	
16	4025	Dispensing state	Start to dispensing	Stopped: 0 Running: 1	Unsigned Short int (2 bytes)	
17	4026	Control mode	Control mode setting	Internal : 0 Footswitch: 1	Unsigned Short int	

				Voltage: 2 Current: 3	(2 bytes)	
18						
19	4028	Slave device address	Slave device address	1-247	Unsigned Short int (2 bytes)	
20	4029	Baud rate	Baud rate setting	4800bbs 0 9600bbs 1 19200bbs 2 38400bbs 3	Unsigned Short int (2 bytes)	
21	4030	External control mode	External control mode setting	Pulse: 0 Level: 1	Unsigned Short int (2 bytes)	
22	4031	Infrared remote setting	Turn infrared remote on or off	Disable: 0 Enable: 1	Unsigned Short int (2 bytes)	
23	4034	Restore default values	Set default values		Unsigned Short int (2 bytes)	
24	4035	Flow rate Coefficient	Flow rate coefficient setting		Float (4 bytes)	
25						
26	4126	Running state	Running state setting	Stop: 0 Start: 1	Unsigned Short int (2 bytes)	
28	4127	MODBUS mode	Switch MODBUS mode	Computer: 0 PLC: 1	Unsigned Short int (2 bytes)	
29	4128	Volume dispensing data structure	Volume dispensing parameter setting	(4128+8) × N) flow rate 4 bytes float (4130+8) × N) volume 4 bytes float (4132+8) × N) downtime 2 bytes unsigned (4133+8) × N) 2nd byte unsigned (4134+8) × N) liquid volume unit 2 bytes, unsigned (1 in microliters, 2 in milliliters, and 3 in liters) (4135+8) × N) flow unit 2 bytes unsigned (uL/min=1, mL/min=2, L/min=3)	16 * 5 bytes, 40 words, N=0-4, a total of five sets of data	
30	4168	Time dispensing data structure	Set time dispensing parameter setting	(4168+6) × N) flow 4 bytes float (4170+6) × N) running time 2 bytes unsigned (4171+6) × N)	12*5bytes 30 words, N=0-4, a total of five sets of data	

				downtime 2 bytes unsigned (4172+6) × N) times 2 byte unsigned (4173+6) × N) flow unit 2 bytes unsigned (uL/min=1, mL/min=2, L/min=3)		
31	4198	Copy dispensing data structure	Set replication dispensing parameter setting	(4198+8) × N) flow 4 bytes floating point (4200+8 × N) total liquid volume 4 bytes floating point (4202+8 × N) downtime 2 bytes unsigned (4203+8 × N) times 2 byte unsigned (4204+8) × N) the unit of total liquid volume is 2 bytes, unsigned (1 in microliters, 2 in milliliters, and 3 in liters) (4205+8 × N) flow unit 2 bytes unsigned (uL/min=1, mL/min=2, L/min=3)	16*5 bytes 40 words, N=0-4, a total of five sets of data	

8.5. Holding Register (Power-off Memory RAM)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	
1	4800	Cumulativ e running times	Record cumulative running times	0- 4294967295	UnSigned Long int (4 bytes)	
2	4802	Cumulativ e running steps	Record cumulative running times (equivalent to cumulative volume)	0- 4294967295	Unsigned Long int (4 bytes)	

9. Parameters and Addresses of Syringe Pump

9.1. Setting and Parameter Address Table for Syringe Pump V1/V2 Series

Product: TYD01-01, TYD01-02, TYD02-01, TYD02-02, TYD02-04, TYD02-10, TYD03-01, TSD01-01,

Add: No.1-11, East Park of Equipment Manufacturing, Yangguang
Street, Xushui District, Baoding City, Hebei, China

Tel: +86-312-6770711

Web: www.leadfluid.com

TFD01-01, TFD02-01, TFD03-01

Mode: RTU
Addresses: 1-247 (User Defined)
Baud Rate: 4800, 9600, 19200, 38400
Data Bits: 8
CRC: Even Parity
Stop Bit: 1

9.1.1. Input Register (not stored)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	
1	(1000)	Internal temperature	Check the temperature inside the pump	-100-+100 Degrees	Signed Short int (2 bytes)	
2	(1001)	Reserved			Unsigned Short int (2 bytes)	
3	(1002)	Infuse speed	Check current Infuse speed	0.001-150rpm	Float (4 bytes)	
4	(1004)	Current running steps	Check current running steps	0-4294967295	Unsigned Long int (4 bytes)	
5	(1006)	Required steps for current steps	Check required steps for current steps	0-4294967295	Unsigned long int (4 bytes)	
6	(1008)	Current running time	Check current running time	0-4294967295	Unsigned long int (4 bytes)	
7	(1012)	Current completion times	Check current completion times	0-4294967295	Unsigned Long int (4 bytes)	
8	(1016)	Value of rotating speed timer	The setting that determines current rotating speed	50-65535	Unsigned Short int (2 bytes)	
9	(1018)	Manufacturer information	Display manufacturer information	"LeadFluid"	Unsigned Char (10 bytes)	
10	(1023)	Product information	Display product information	"TYD01"	Unsigned Char (10 bytes)	
11	(1028)	Touch panel X-coordinate	Display touch panel X-coordinate		Unsigned Short int (2 bytes)	
12	(1029)	Touch panel Y-coordinate	Display touch panel Y-coordinate		Unsigned Short int (2 bytes)	
13						
14	1032	Cumulative volume	Checking the cumulative volume		Float (4 bytes)	
15	1034	Unit of cumulative	Check unit of cumulative volume	nL: 0 μL: 1	Unsigned Short int	

		volume		mL: 2 L: 3	(2 bytes)	
16	1035	Consumption of liquid	Checking the value of liquid consumption		Float (4 bytes)	
17	1037	Liquid consumption unit	Checking liquid consumption unit	nL: 0 μL: 1 mL: 2	Unsigned Short int (2 bytes)	
18	1038	Remaining liquid volume	Checking residual liquid volume		Float (4 bytes)	
19	1040	Remaining liquid unit	Checking liquid volume unit	nL: 0 μL: 1 mL: 2	Unsigned Short int (2 bytes)	
20	1041	Withdraw speed	Checking withdraw speed	0.001-150rpm	Float (4 bytes)	
21	1043	Time consuming	Checking consumption time	0-4294967295 Unit: milliseconds	Unsigned Long int (4 bytes)	
2	1045	Remainder	Checking remaining time	0-4294967295 Unit: milliseconds	Unsigned Long int (4 bytes)	
23	1047	Alarm state	Checking for alarms	Normal: 0 Alarm: 1	Unsigned Short int (2 bytes)	
24	1052	Maximum flow rate	The maximum flow rate of the current syringe	Unit: nL/min	Float (4 bytes)	Suitable for software versions 1.15 and above
25	1054	Minimum flow rate	The minimum flow rate of the current syringe	Unit: nL/min	Float (4 bytes)	Suitable for software versions 1.15 and above
26	1056	Maximum flow rate unit	The maximum flow rate unit of the current syringe	nL: 0 μL: 1 mL: 2	Unsigned Short int (2 bytes)	
27	1057	Minimum flow rate	The minimum flow rate unit of the current syringe	nL: 0 μL: 1 mL: 2	Unsigned Short int (2 bytes)	

9.1.2. Input Register (Not Stored)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	
1	(2800)	Total on time	Check cumulative time that the pump powered on	0-4294967295 Unit: second	Unsigned Long int (4 bytes)	
2	(2802)	Total running time	Check cumulative running time	0-4294967295 Unit: second	Unsigned Long int (4 bytes)	
3	(2804)	Total powered on cycles	Check the cycles the pump has powered on	0-4294967295	Unsigned Long int (3 bytes)	

9.1.3. Holding Register (Not Stored)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	
1*	(3001)	Fast infuse	Start or stop quick infuse	Stopped: 0 Fast forward: 1	Unsigned Short int (2 bytes)	
2*	(3002)	Fast withdraw	Start or stop quick withdraw	Stopped: 0 Fast forward: 1	Unsigned Short int (2 bytes)	

9.1.4. Holding Register (Power-off Memory FRAM)

No.	Address	Name	Function	Data Range	Data Type	
1	4000	The leftmost physical coordinate of touch screen	The leftmost physical coordinate of touch screen		Unsigned Long int (4 bytes)	
2	4002	The rightmost physical coordinate of touch screen	The rightmost physical coordinate of touch screen		Unsigned Long int (4 bytes)	
3	4004	The uppermost physical coordinate of touch screen	The uppermost physical coordinate of touch screen		Unsigned Long int (4 bytes)	
4	4006	The down most physical coordinate of touch screen	The down most physical coordinate of touch screen		Unsigned Long int (4 bytes)	
5	4008	Number of pre stored parameter groups	Select the number of groups for pre stored parameters	Group 1: 0 Group 2: 1 Group 3: 2	Unsigned Short int (2 bytes)	
6	4015	Flow rate	Flow rate in continuous mode		Float (4 bytes)	
7	4017	Working mode	Working mode setting	Only infuse: 0 Only withdraw: 1 Withdraw/infuse: 2 Infuse/withdraw: 3 Continuous: 4	Unsigned Short int (2 bytes)	
8	4018	Key tone	Key tone setting	Tone off: 0 Tone on: 1	Unsigned Short int (2 bytes)	
9	4019	Parameter lock	Parameter lock setting	Unlock: 0 Lock: 1	Unsigned Short int	

					(2 bytes)	
10	4020	Language	Language setting	Chinese: 0 English: 1	Unsigned Short int (2 bytes)	
11	4021	Syringe number	Syringe number		Unsigned Short int (2 bytes)	
12	4022	Flow rate units	Flow rate units setting for continuous mode	nL/min: 0 uL/ min: 1 mL/ min: 2	Unsigned Short int (2 bytes)	
13	4023	Direction of movement	Current movement direction (can be set in continuous mode)	Withdraw direction: 0 Infuse direction: 1	Unsigned Short int (2 bytes)	
14	4024	Paused state	Pause state setting	Terminate: 0 Pause: 1	Unsigned Short int (2 bytes)	
15	4025	Dispensing state	Start to dispense	Stop: 0 Start/Continue: 1	Unsigned Short int (2 bytes)	
16	4026	Control mode	Set external control signal control method: pulse or level (can be set in continuous mode)	Pulse: 0 Level: 1	Unsigned Short int (2 bytes)	
17	4027	Push force	Push force settling	1-100	Unsigned Short int (2 bytes)	
18	4028	Slave address	Slave address setting	1-247	Unsigned Short int (2 bytes)	
19	4029	Communication baud rate	Set the number of communication baud rate	4800bps 0 9600bps 1 19200bps 2 38400bps 3	Unsigned Short int (2 bytes)	
20	4030	LCD backlight	Set the brightness of the LCD backlight	20-100	Unsigned Short int (2 bytes)	
21	4035	Velocity coefficient	Set the coefficient for custom syringes		Float (4 bytes)	
22	4034	Restore default values	Set default values	Restore: 55 Normal: 170	Unsigned Short int (2 bytes)	
23	4087	Traffic alarm	Traffic alarm setting	Turn on: 0 Turn off: 1	Unsigned Short int (2 bytes)	
24	4088	Customized syringe ID	Customized syringe ID setting	0.001-40mm	Float (4 bytes)	
25	4090	Customized syringe volume	Customized syringe volume setting	0.001-200	Float (4 bytes)	
26	4092	Custom syringe unit	Set the volume unit of a custom syringe	uL: 1 mL: 2	Unsigned Short int (2 bytes)	
27	4093	Hours of interval time	Hours for interval time setting	0-999	Unsigned Short int	

					(2 bytes)	
28	4094	Minutes of interval time	Minutes for interval time setting	0-59	Unsigned Short int (2 bytes)	
29	4095	Seconds for interval time	Seconds for interval time setting	0-59	Unsigned Short int (2 bytes)	
30	4096	Milliseconds for interval time	Milliseconds for interval time setting	0-99	Unsigned Short int (2 bytes)	
31	4097	Number of cycles	Set the number of cycles	0-999	Unsigned Short int (2 bytes)	
32	4126	Running state	Set whether to run or stop		Unsigned Short int (2 bytes)	
33	4127	MODBUS mode	Switch MODBUS mode	Computer or PLC	Unsigned Short int (2 bytes)	
34	4128	Only infuse data structure	Set parameters for infuse only (three sets in total)	4128+6 * N liquid volume 4 bytes float 4130+6*N fluid infuse 4 bytes float 4132+6 * N liquid volume unit 2 bytes unsigned 4133+6*N infuse flow unit 2 bytes unsigned	12*3 bytes 18 words, N=0-2, three sets of data in total	
35	4146	Only withdraw data structure	Set only withdraw parameters (three sets in total)	4146+6*N liquid volume 4 bytes float 4148+6*N withdraw flow 4 bytes float 4150+6*N volume unit 2 bytes unsigned 4151+6*N withdraw flow unit 2 bytes unsigned	12*3 bytes 18 words, N=0-2, three sets of data in total	
36	4164	Withdraw data structure	Set withdraw parameters (three sets in total)	4164+10*N liquid volume 4 bytes float 4166+10*N infuse flow 4 bytes float 4168+10*N withdraw flow 4 bytes floating point 4170+10*N volume unit 2 bytes unsigned	20*3 bytes 30 words, N=0-2, three sets of data in total	

				4171+10*N infuse flow unit 2 bytes unsigned 4172+10*N withdraw flow unit 2 bytes unsigned		
37	4194	Infuse withdraw data structure	Set infuse withdraw parameters (three sets in total)	4194+10*N liquid volume 4 bytes float 4196+10*N infuse flow 4 bytes float 4198+10*N withdraw flow 4 bytes float 4200+10*N volume unit 2 bytes unsigned 4201+9*N flow unit 2 bytes unsigned 4202+10*N withdraw flow unit 2 bytes unsigned	20 * 3 bytes, 30 words, N=0-2, three sets of data in total	

9.1.5. Holding Register (Power-off Memory RAM)

No	Address	Name	Function	Data Range	Data Type	
1	4800	Total number of running times	Record cumulative running times	0- 4294967295	UnSigned Long int (4 bytes)	
2	4802	Total number of running steps	Record cumulative running steps (equivalent to cumulative volume)	0- 4294967295	Unsigned Long int (4 bytes)	

9.2. Setting and Parameter Address Table for Syringe Pump V3 Series

Product: TYD01-01-V3, TYD01-02-V3, TYD02-01-V3, TYD02-02-V3, TYD02-04-V3, TYD02-10-V3, TYD03-01-V3, TSD01-01-V3, TFD01-01-V3, TFD02-01-V3, TFD03-01-V3

Mode: RTU
Addresses: 1-247 (User Defined)
Baud Rate: 4800, 9600, 19200, 38400
Data Bits: 8
CRC: Even Parity
Stop Bit: 1

9.2.1. Input Register (not stored)

No	Address	Name	Function	Data Range	Data Type	
----	---------	------	----------	------------	-----------	--

.						
1	(1000)	Internal temperature	Check the temperature inside the pump	-100-+100 Degrees	Signed Short int (2 bytes)	
2	(1001)	Reserved			Unsigned Short int (2 bytes)	
3	(1002)	Infuse speed	Check current Infuse speed	0.001-150rpm	Float (4 bytes)	
4	(1004)	Required steps for current steps	Check required steps for current steps	0-4294967295	Unsigned Long int (4 bytes)	
5	(1006)	Remaining running steps	Check the remaining running steps	0-4294967295	Unsigned long int (4 bytes)	
6	(1008)	Current running time	Check current running time Unit:10 ms	0-4294967295	Unsigned long int (4 bytes)	
7	(1012)	Current completion times	Check current completion times	0-4294967295	Unsigned Long int (4 bytes)	
8	(1016)	Value of rotating speed timer	The setting that determines current rotating speed	50-65535	Unsigned Short int (2 bytes)	
9	(1018)	Manufacturer information	Display manufacturer information	"LeadFluid"	Unsigned Char (10 bytes)	
10	(1023)	Product information	Display product information	"TYD01"	Unsigned Char (10 bytes)	
11	1032	Cumulative volume	Checking the cumulative volume		Float (4 bytes)	
12	1034	Unit of cumulative volume	Check unit of cumulative volume	nL: 0 μL: 1 mL: 2 L: 3	Unsigned Short int (2 bytes)	
13	1035	Consumption of liquid	Checking the value of liquid consumption		Float (4 bytes)	
14	1037	Liquid consumption unit	Checking liquid consumption unit	nL: 0 μL: 1 mL: 2	Unsigned Short int (2 bytes)	
15	1038	Remaining liquid volume	Checking residual liquid volume		Float (4 bytes)	
16	1040	Remaining liquid unit	Checking liquid volume unit	nL: 0 μL: 1 mL: 2	Unsigned Short int (2 bytes)	
17	1043	Time consuming	Checking consumption time	0-4294967295 Unit: milliseconds	Unsigned Long int (4 bytes)	
18	1045	Remainder	Checking remaining time	0-4294967295 Unit: milliseconds	Unsigned Long int (4 bytes)	
19	1047	Alarm state	Checking for alarms	Normal: 0 Alarm: 1	Unsigned Short int	

					(2 bytes)	
20	1052	Maximum flow rate	The maximum flow rate of the current syringe		Float (4 bytes)	
21	1054	Minimum flow rate	The minimum flow rate of the current syringe	Unit: nL/min	Float (4 bytes)	
22	1056	Maximum flow rate unit	The maximum flow rate unit of the current syringe	nL: 0 μL: 1 mL: 2	Unsigned Long int (2 bytes)	
23	1057	Minimum flow rate unit	The minimum flow rate unit of the current syringe	nL: 0 μL: 1 mL: 2	Unsigned Long int (2 bytes)	

9.2.2. Input Register (Not Stored)

No.	Address	Name	Function	Data Range	Data Type	
1	(2800)	Total on time	Check cumulative time that the pump powered on	0-4294967295 Unit: second	Unsigned Long int (4 bytes)	
2	(2802)	Total running time	Check cumulative running time	0-4294967295 Unit: second	Unsigned Long int (4 bytes)	
3	(2804)	Total powered on cycles	Check the cycles the pump has powered on	0-4294967295	Unsigned Long int (3 bytes)	

9.2.3. Holding Register (Not Stored)

No.	Address (Decimal)	Name	Function	Data Range	Data Type	
1	(3001)	Fast infuse	Start or stop quick infuse	Stopped: 0 Fast forward: 1	Unsigned Short int (2 bytes)	
2	(3002)	Fast withdraw	Start or stop quick withdraw	Stopped: 0 Fast forward: 1	Unsigned Short int (2 bytes)	

9.2.4. Holding Register (Power-off Memory FRAM)

No.	Address	Name	Function	Data Range	Data Type	
1	4008	Number of pre stored parameter groups	Select the number of groups for pre stored parameters	Group 1: 0 Group 2: 1 Group 3: 2	Unsigned Short int (2 bytes)	
2	4015	Flow rate	Flow rate in continuous mode		Float (4 bytes)	

3	4017	Working mode	Working mode setting	Only infuse: 0 Only withdraw: 1 Withdraw/infuse: 2 Infuse/withdraw: 3 Continuous: 4	Unsigned Short int (2 bytes)	
4	4018	Key tone	Key tone setting	Tone on: 1 Tone off: 0	Unsigned Short int (2 bytes)	
5	4019	Parameter lock	Parameter lock setting	Unlock: 0 Lock: 1	Unsigned Short int (2 bytes)	
6	4020	Language	Language setting	Chinese: 0 English: 1	Unsigned Short int (2 bytes)	
7	4021	Syringe number	Syringe number		Unsigned Short int (2 bytes)	
8	4022	Flow rate units	Flow rate units setting for continuous mode	nL/min: 0 uL/ min: 1 mL/ min: 2	Unsigned Short int (2 bytes)	
9	4023	Direction of movement	Current movement direction (In continuous mode, the pulse mode can be configured; in level mode, it cannot be configured)	Withdraw direction: 0 Infuse direction: 1	Unsigned Short int (2 bytes)	
10	4024	Paused state	Pause state setting	Continue: 0 Pause: 1	Unsigned Short int (2 bytes)	
11	4025	Dispensing state	Start to dispense	Stop: 0 Start/Continue: 1	Unsigned Short int (2 bytes)	
12	4026	Control mode	Set external control signal control method: pulse or level (can be set in continuous mode)	Pulse: 0 Level: 1	Unsigned Short int (2 bytes)	
13	4027	Push force	Push force settling	25%-100%	Unsigned Short int (2 bytes)	
14	4028	Slave address	Slave address setting	1-247	Unsigned Short int (2 bytes)	
15	4029	Communication baud rate	Set the number of communication baud rate	4800bps 0 9600bps 1 19200bps 2 38400bps 3	Unsigned Short int (2 bytes)	
16	4030	LCD backlight	Set the brightness of the LCD backlight	20-100	Unsigned Short int (2 bytes)	
17	4035	Velocity coefficient	Set the coefficient for custom		Float (4 bytes)	

			syringes			
18	4034	Restore default values	Set default values	Restore: 55 Normal: 170	Unsigned Short int (2 bytes)	
19	4087	Traffic alarm	Traffic alarm setting	Turn on: 0 Turn off: 1	Unsigned Short int (2 bytes)	
20	4088	Customized syringe ID	Customized syringe ID setting	0.001-40mm	Float (4 bytes)	
21	4090	Customized syringe volume	Customized syringe volume setting	0.001-200mL	Float (4 bytes)	
22	4093	Hours of interval time	Hours for interval time setting	0-999	Unsigned Short int (2 bytes)	
23	4094	Minutes of interval time	Minutes for interval time setting	0-59	Unsigned Short int (2 bytes)	
24	4095	Seconds for interval time	Seconds for interval time setting	0-59	Unsigned Short int (2 bytes)	
25	4096	Milliseconds for interval time	Milliseconds for interval time setting	0-99	Unsigned Short int (2 bytes)	
26	4097	Number of cycles	Set the number of cycles	0-999	Unsigned Short int (2 bytes)	
27	4126	Continuous Running State	Set whether to run or stop in continuous mode	Stopped:0 Runing:1	Unsigned Short int (2 bytes)	
28	4127	MODBUS mode	Switch MODBUS mode	Computer or PLC(CDAB or ABCD)	Unsigned Short int (2 bytes)	
29	4128	Only infuse data structure	Set parameters for infuse only (three sets in total)	4128+6 * N liquid volume 4 bytes float 4130+6*N fluid infuse4 bytes float 4132+6 * N liquid volume unit 2 bytes unsigned 4133+6*N infuse flow uniy2 bytes unsigned	12*3 bytes 18 words, N=0-2, three sets of data in total	
30	4146	Only withdraw data structure	Set only withdraw parameters (three sets in total)	4146+6*N liquid volume 4 bytes float 4148+6*N withdraw flow 4 bytes float 4150+6*N volume unit 2 bytes unsigned 4151+6*N withdraw flow	12*3 bytes 18 words, N=0-2, three sets of data in total	

				unit 2 bytes unsigned		
31	4164	Withdraw data structure	Set withdraw parameters (three sets in total)	4164+10*N liquid volume 4 bytes float 4166+10*N infuse flow 4 bytes float 4168+10*N withdraw flow 4 bytes floating point 4170+10*N volume unit 2 bytes unsigned 4171+10*N infuse flow unit 2 bytes unsigned 4172+10*N withdraw flow unit 2 bytes unsigned	20*3 bytes 30 words, N=0-2, three sets of data in total	
32	4194	Infuse withdraw data structure	Set infuse withdraw parameters (three sets in total)	4194+10*N liquid volume 4 bytes float 4196+10*N infuse flow 4 bytes float 4198+10*N withdraw flow 4 bytes float 4200+10*N volume unit 2 bytes unsigned 4201+9*N flow unit 2 bytes unsigned 4202+10*N withdraw flow unit 2 bytes unsigned	20 * 3 bytes, 30 words, N=0-2, three sets of data in total	
33	4224	Only infuse the time data structure	Set only infuse time unit mode parameters (three sets in total)	4224+4*N infuse time 4 bytes float 4226+4*N infuse time unit 2 bytes unsigned	8*3 bytes 12 words N=0-2, three sets of data in total	
34	4236	Only withdraw the time data structure	Set only withdraw time unit mode parameters (three sets in total)	4236+4*N withdraw time 4 bytes float 4238+4*N withdraw time unit 22 bytes unsigned	8*3 bytes 12 words N=0-2, three sets of data in total	
35	4248	Withdraw/ infuse the time data	Set only withdraw/infuse time unit mode	4248+6*N infuse time 4 bytes float 4250+6*N	8*3 bytes 18 words N=0-2,	

		structure	parameters (three sets in total)	withdraw time 4 bytes float 4252+6*N infuse time unit 22 bytes unsigned 4253+6*N withdraw time unit 2 bytes unsigned	three sets of data in total	
36	4266	Infuse withdraw the time data structure	Set only infuse/withdraw time unit mode parameters (three sets in total)	4266+6*N infuse time 4 bytes float 4268+6*N withdraw time 4 bytes float 4270+6*N infuse time 22 bytes unsigned 4271+6*N withdraw time unit 22 bytes unsigned	8*3 bytes 18 words N=0-2, three sets of data in total	

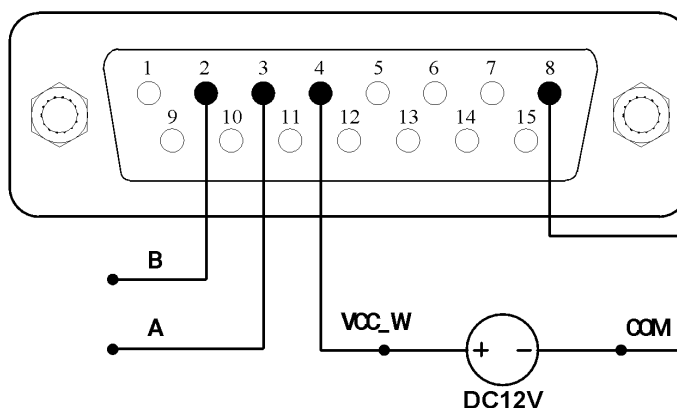
9.2.5. Holding Register (Power-off Memory RAM)

No	Address	Name	Function	Data Range	Data Type	
1	4800	Total number of running times	Record cumulative running times	0- 4294967295	UnSigned Long int (4 bytes)	
2	4802	Total number of running steps	Record cumulative running steps (equivalent to cumulative volume)	0- 4294967295	Unsigned Long int 10. bytes)	

10. RS485 hardware interface (DB15 pin) for Lead Fluid products

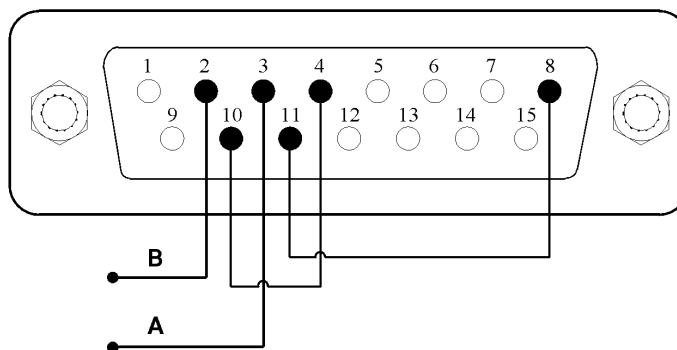
(1) External power supply

Connect the positive end of the external+24V or +12V power supply to the 4-pin VCC_ W connection, connect the negative end with 8-pin COM



(2) Internal power supply

Connect 10-pin+12V and 4-pin VCC of DB15-pin_ W connection, connect 11-pin GND with 8-pin COM



11.Example of Communication Protocol Data Frame

11.1. Read Input Register

Temperature: address 1000, L or F series, 1 word length, integer variable

000000-Tx: 01 04 03 E8 00 01 B1 BA

000001-Rx: 01 04 02 00 22 39 29

Read

	HEX
Slave address	01
Function code	04
Register High	03
Register low	E8
Number of registers high	00
Number of registers low	01
CRC high bit	B1
CRC low bit	BA

Description: Address 1000(03E8)

Response

	HEX
Slave address	01
Function code	04
Number of bytes	02
Data-high	00
Data-low	22
CRC high bit	39
CRC low bit	29

Description: actual return value 34(00 22)

11.2. Read and Keep Registers

Flow: Address 4015, L or F series, 2 words length, single precision floating point variable, byte order CDAB (original computer mode)

000032-Tx:01 03 0F AF 00 02 F7 3E

000033-Rx:01 03 04 EF CA 42 65 1E 52

Read

	HEX
Slave address	01
Function code	03
Register High	0F
Register Low	AF

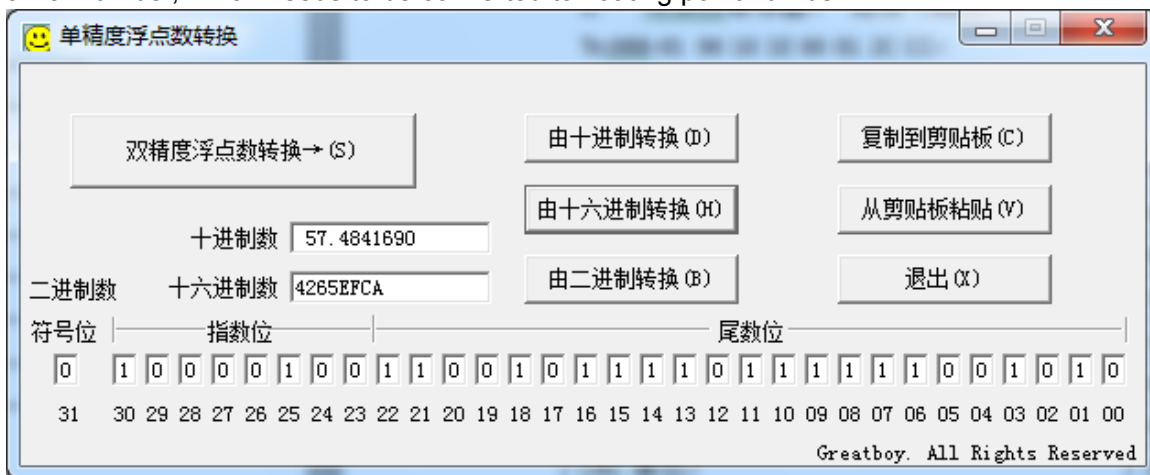
Number of registers high	00
Number of registers low	02
CRC high bit	F7
CRC low bit	3E

Description: Address 4015 (0F AF)

Response

	HEX
Slave address	01
Function code	03
Number of bytes	04
Data 1 high bit (4015)	EF
Data 1 low bit (4015)	CA
Data 2 high bit (4016)	42
Data 2 low bit (4016)	65
CRC high bit	1E
CRC low bit	52

Description: The actual return value is 57.484169 (42 65 EF CA). The byte order selected here is CDAB (original communication mode: computer mode), and the data format is data 2+data 1 to form a four-byte hexadecimal number, which needs to be converted to floating point number



11.3. Write Holding Register

Start / stop: Address 4126, L or F series, 1 word length, integer variable

Tx:088-01 06 10 1E 00 01 2C CC

Rx:089-01 06 10 1E 00 01 2C CC

Write

	HEX
Slave address	01
Function code	06
Register High	10
Register Low	1E
Number of registers high	00
Number of registers low	01
CRC high bit	2C
CRC low bit	CC

Description: Address 4126 (10 1E)

Response

	HEX
Slave address	01
Function code	06
Register High	10
Register Low	1E

Number of registers high	00
Number of registers low	01
CRC high bit	CC
CRC low bit	CC

11.4. Write Holding Register

Flow: address 4015, L or F series, 2 word length, single-precision floating point variable, byte order CDAB (original computer mode)

Tx:000082-01 10 0F AF 00 02 04 00 00 41 20 C9 EF

Rx:000083-01 10 0F AF 00 02 72 FD

Write

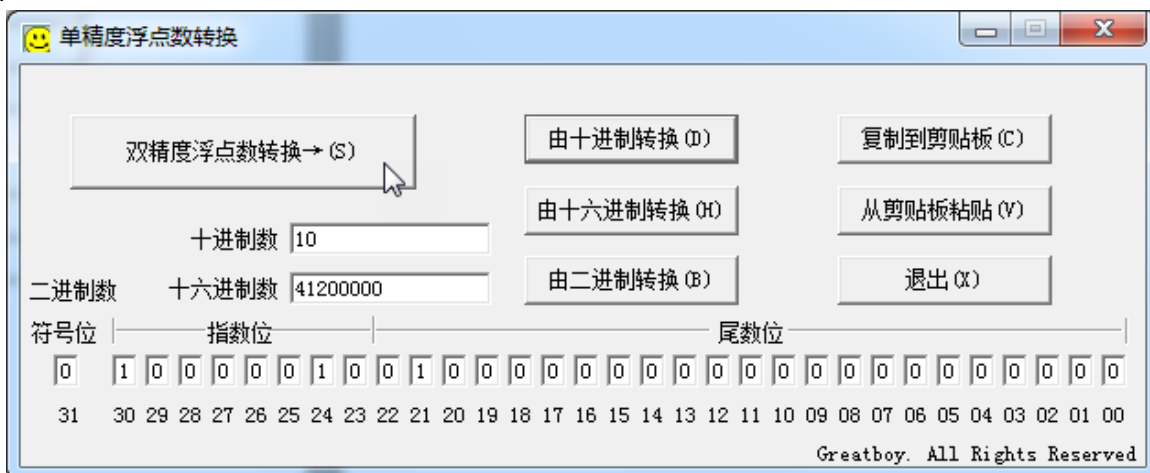
	HEX
Slave address	01
Function code	10
Register High	0F
Register Low	AF
Number of registers high	00
Number of registers low	02
Number of bytes	04
Data 1 high bit (4015)	00
Data 1 low bit (4015)	00
Data 2 high bit (4016)	41
Data 2 low bit (4016)	20
Slave address	C9
Function code	EF

Description: Address 4015 (0F AF). Data 10.0 (41 20 00 00)

Response

	HEX
Slave address	01
Function code	10
Register High	0F
Register Low	AF
Number of registers high	00
Number of registers low	02
CRC high bit	72
CRC low bit	FD

Note: The byte order selected here is CDAB (original communication mode: computer mode), and the data format is data 2+data 1 to form a four-byte hexadecimal number, which needs to be converted to floating point number



11.5. Read holding register

Flow: address 4015, L or F series, 2 word length, single-precision floating point variable, byte order ABCD (original PLC mode)

000032-Tx:01 03 0F AF 00 02 F7 3E

000033-Rx:01 03 04 42 65 EF CA 32 33

Read

	HEX
Slave address	01
Function code	03
Register High	0F
Register Low	AF
Number of registers high	00
Number of registers low	02
CRC high bit	F7
CRC low bit	3E

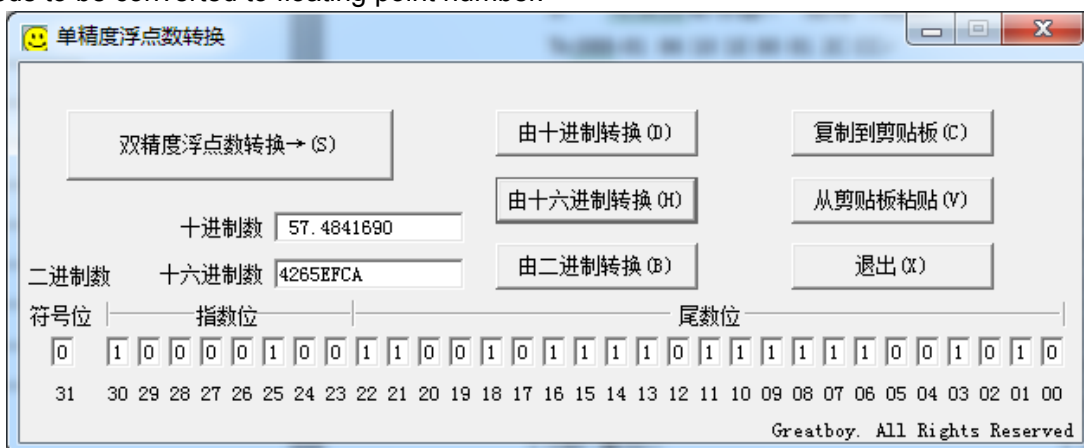
Description: Address4015(0F AF)

Response

	HEX
Slave address	01
Function code	03
Number of bytes	04
Data 1 high bit (4015)	42
Data 1 low bit (4015)	65
Data 2 high bit (4016)	EF
Data 2 low bit (4016)	CA
CRC high bit	32
CRC low bit	33

Description: The actual return value is 57.484169 (42 65 EF CA).

The byte order selected here is ABCD (original communication mode: PLC mode). Its data sorting order is different from the computer mode. The data format is data 1+data 2 to form a four-byte hexadecimal number, which needs to be converted to floating point number.



12.Appendix 1: Solution of hexadecimal to single-precision floating point number

```
Union
{
float f
char buf[4]
```

```
void write_Df(u16 ndata, float df)
{
    u16  d0,d1
    data.f = df

    d0 = (data.buf [1] << 8) + data.buf [0] ;
    d1 = (data.buf [3] << 8) + data.buf [2] ;
}
```

```
0x11, 0xD1, 0xD0, 0x10, 0xF0, 0x30, 0x31, 0xF1, 0x33, 0xF3, 0xF2, 0x32, 0x36, 0xF6, 0xF7,  
0x37, 0xF5, 0x35, 0x34, 0xF4, 0x3C, 0xFC, 0xFD, 0x3D, 0xFF, 0x3F, 0x3E, 0xFE, 0xFA, 0x3A,  
0x3B, 0xFB, 0x39, 0xF9, 0xF8, 0x38, 0x28, 0xE8, 0xE9, 0x29, 0xEB, 0x2B, 0x2A, 0xEA, 0xEE,  
0x2E, 0x2F, 0xEF, 0x2D, 0xED, 0xEC, 0x2C, 0xE4, 0x24, 0x25, 0xE5, 0x27, 0xE7, 0xE6, 0x26,  
0x22, 0xE2, 0xE3, 0x23, 0xE1, 0x21, 0x20, 0xE0, 0xA0, 0x60, 0x61, 0xA1, 0x63, 0xA3, 0xA2,  
0x62, 0x66, 0xA6, 0xA7, 0x67, 0xA5, 0x65, 0x64, 0xA4, 0x6C, 0xAC, 0xAD, 0x6D, 0xAF, 0x6F,  
0x6E, 0xAE, 0xAA, 0x6A, 0x6B, 0xAB, 0x69, 0xA9, 0xA8, 0x68, 0x78, 0xB8, 0xB9, 0x79, 0xBB,  
0x7B, 0x7A, 0xBA, 0xBE, 0x7E, 0x7F, 0xBF, 0x7D, 0xBD, 0xBC, 0x7C, 0xB4, 0x74, 0x75, 0xB5,  
0x77, 0xB7, 0xB6, 0x76, 0x72, 0xB2, 0xB3, 0x73, 0xB1, 0x71, 0x70, 0xB0, 0x50, 0x90, 0x91,  
0x51, 0x93, 0x53, 0x52, 0x92, 0x96, 0x56, 0x57, 0x97, 0x55, 0x95, 0x94, 0x54, 0x9C, 0x5C,  
0x5D, 0x9D, 0x5F, 0x9F, 0x9E, 0x5E, 0x5A, 0x9A, 0x9B, 0x5B, 0x99, 0x59, 0x58, 0x98, 0x88,  
0x48, 0x49, 0x89, 0x4B, 0x8B, 0x8A, 0x4A, 0x4E, 0x8E, 0x8F, 0x4F, 0x8D, 0x4D, 0x4C, 0x8C,  
0x44, 0x84, 0x85, 0x45, 0x87, 0x47, 0x46, 0x86, 0x82, 0x42, 0x43, 0x83, 0x41, 0x81, 0x80,  
0x40  
}
```